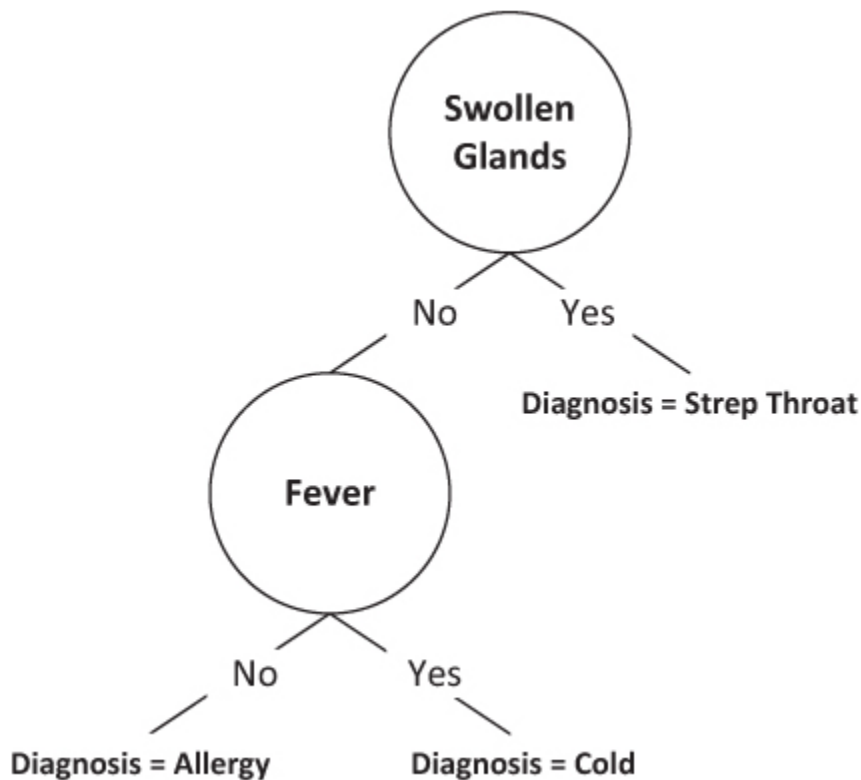


Practice Exercise: Diagnosing Illness (Decision Trees)

1. Use the decision tree to classify patients
2. Modify the production rules by adding a new attribute
3. Draw the updated decision tree by hand
4. Submit your completed work

1. Test the model on new patients



Use the decision tree diagram to classify the patients below.
Write your diagnosis in the final column.

Patient	Swollen Glands	Fever	Diagnosis
14	Yes	No	
15	No	No	
16	No	Yes	
17	Yes	Yes	

2. Modify the production rules

To improve the diagnosis model, add a new attribute: duration of symptoms:

- Short: less than 3 days

- Long: 3 days or more

Use this attribute to refine each diagnosis as either acute (short duration) or chronic (long duration).

Original production rules:

IF swollen glands = YES THEN diagnosis = strep throat

IF swollen glands = NO AND fever = YES THEN diagnosis = cold

IF swollen glands = NO AND fever = NO → allergy

Your task: Write a complete new set of production rules that incorporates the duration attribute. Each original rule will split into two rules (for short/long duration).

Required: Every original rule must be expanded to include duration. Each final rule must include all relevant conditions. Each diagnosis must be labeled as either acute or chronic.

Write your updated rules below:

3. Draw the new decision tree

Draw your updated decision tree below. Use the original diagram as a reference.